Friendship Prediction on Twitch Using Metadata and Graph Features

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Abstract

Twitch has evolved to be a platform where users establish communities and relations beyond live streaming. Friendship prediction in such environments can give recommendations and increase engagement of users. This research addresses the task of friendship prediction between Twitch users by integrating metadata as well as graph-based features. Metadata contains language, affiliate status, maturity flags, and lifetime of an account, whereas graph features extract relationship information by computing metrics such as Common Neighbors, Jaccard similarity, Preferential Attachment, and Adamic-Adar. We train conventional models like Random Forest, SVM, XGBoost, and Logistic Regression, and utilize Node2Vec embeddings to learn network structure. Experiments demonstrate that using graph-based features improves model performance drastically in contrast to metadata only. Between all tested models, combining Node2Vec and Adamic-Adar provides maximum accuracy and AUC values, reflecting that social structure is of maximum impact in predicting friendship. The results highlight the usefulness of network-aware approaches and provide promising avenues for scalable link prediction in large social graphs.

1 Introduction

Twitch, initially designed as a streaming site focusing on live gameplay, has turned into an active social space where users group together, subscribe to and follow other users, and interact with each other regularly [1]. As both the number of users and interaction dynamics grow, knowing and forecasting social relationships becomes ever more significant. Friendship prediction—the activity of predicting whether users have or will develop any relationship between them—can be vital in improving users' engagement, enhancing recommender systems, and extracting community structures in such networks.

Even with access to such user metadata as language, affiliate status, and account age, these factors by themselves tend to be insufficient to model social interactions accurately [2]. By focusing only on these profile-level features and ignoring the structure of the network, traditional machine learning models do not account for such structural features as shared neighbors and mutual friends, known to be effective measures of social closeness in previous works.

This work investigates friendship prediction on Twitch by combining both user metadata and graph-based features. We derive features both at the individual level and at the level of network structure, and compare model performances of Random Forest, Logistic Regression, SVM, and embedding-based models with Node2Vec and XGBoost. We then analyze how varying subgraph sampling procedures impact the learned embedding quality.

Our results indicate that integrating graph-theoretic features greatly improves prediction accuracy and resilience over metadata-only models. The combination of Node2Vec embeddings and Adamic-Adar scores provides near-optimal performance, emphasizing global and local network structure. Our results establish that graph-aware machine learning approaches can be valuable tools for link prediction in massive-scale social platforms such as Twitch.

2 Related Work

Heuristic link prediction algorithms have traditionally played the role of simple, effective baselines in network analysis. Such algorithms calculate heuristic similarity between two nodes purely based on graph topology, and no training. Examples range from *Common Neighbors (CN)*, counting common neighbors of two nodes; to *Jaccard coefficient*, normalizing by size of total neighbor set; and *Adamic–Adar (AA) index*, assigning higher weights to rarer common neighbors [2]. Such metrics follow the intuition that two nodes with higher mutual contacts or common features in the network tend to form a link. More sophisticated heuristic indices like *Preferential Attachment* (the degree product) apply this intuition to prefer popular nodes. Such heuristic algorithms (similarity-based or topological indices) are "hand-crafted" as they consist of predefined formulae based on network structure and not learned from data. Such heuristic algorithms provide a low-cost, interpretable link prediction baseline and tend to perform well surprisingly in friendship networks [2], ignoring node features.

Unlike using just one structural heuristic, another line of research frames link prediction as a *supervised learning* task with hand-engineered features (also known as handcrafted features). Here, every candidate pair of nodes is represented by a feature vector that encodes different aspects of the pair (e.g., profile metadata similarity and structural measures), and a binary predictor is learned to separate real friendships from pairs of unconnected nodes as discussed by Cukierski *et al.* [3]. This formulation enables one to extract many different predictors—like common friend count, whether two users share the same language, or account age difference—as features. Cukierski *et al.* showcase this idea by combining 94 different graph features (variations of CN, AA, and other indices) in a Random Forest predictor and achieve significantly higher link prediction accuracy in a large social network. Likewise, other research has observed that an ensemble of handcrafted features performs strictly better than any individual index: e.g., algorithms like logistic regression or boosting can average multiple link predictors (CN, Jaccard, AA, etc.) to gain higher precision. Backstrom and Leskovec's *Supervised Random Walks* is an exemplar that unites network structure with node properties in an integrated model [4]. Their algorithm learns to skew random walks on the friendship network with node and edge features (like user profile and interaction frequency) as input, and successfully hybridizes structural closeness with attribute similarity to forecast novel friendships

Beyond manual feature design, recent work has looked to *representation learning* to learn link prediction features automatically. Network embedding algorithms learn low-dimensional node vector representations encoding network structure that can be used to predict missing links [6]. One notable example is *Node2Vec*, where node embeddings are learned by simulating biased random walks and using a skip-gram objective to conserve network neighborhoods. Grover and Leskovec demonstrated that using Node2Vec embeddings to predict links outperforms classical heuristics by far on benchmark graphs [5]. Such embeddings encode higher-order patterns of connectivity (community membership, structural role, etc.) implicitly without needing explicit feature design. Other representation learning methods incorporate matrix factorization and deep graph neural networks that learn to predict links directly. For example, graph autoencoder models and graph convolutional networks can embed nodes and optimize link reconstruction as well, enhancing accuracy in many cases. Such representation learning algorithms consequently automate feature extraction from the graph. But they tend to be more expensive and harder to interpret, so in practice, they may be used alongside simpler graph features to trade off efficiency and interpretability. Nonetheless, learned embeddings have driven state-of-the-art results in link prediction, particularly in large-scale social networks where dense structural patterns exist.

3 Methodology

This section describes the methodology used to predict friendships between Twitch users by leveraging both user metadata and graph-based features. The overall pipeline consists of data preprocessing, feature engineering, model training, and evaluation.

3.1 Dataset Overview

We used the Twitch Gamers Dataset [7], which includes two components: a list of undirected edges representing friendships (`large_twitch_edges.csv`) and a metadata file containing user attributes (`large_twitch_features.csv`). The dataset contains 168,114 nodes and 6.8

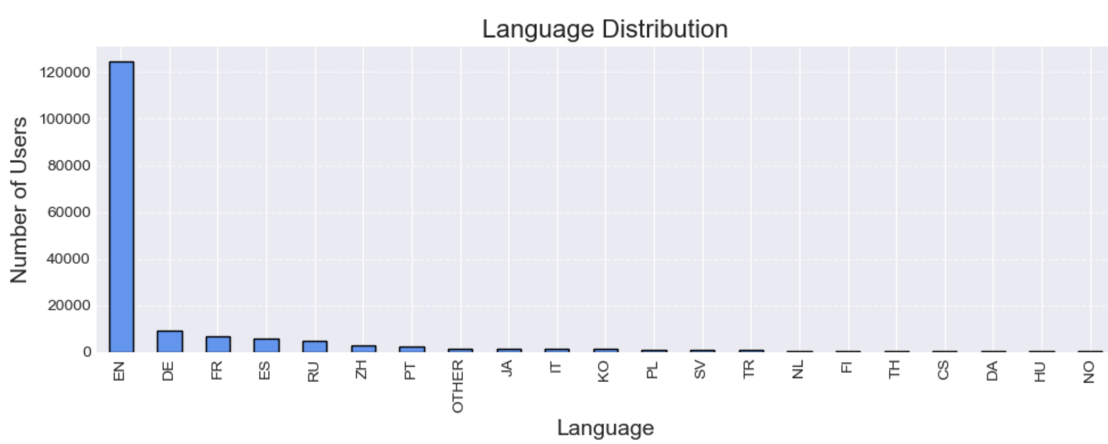
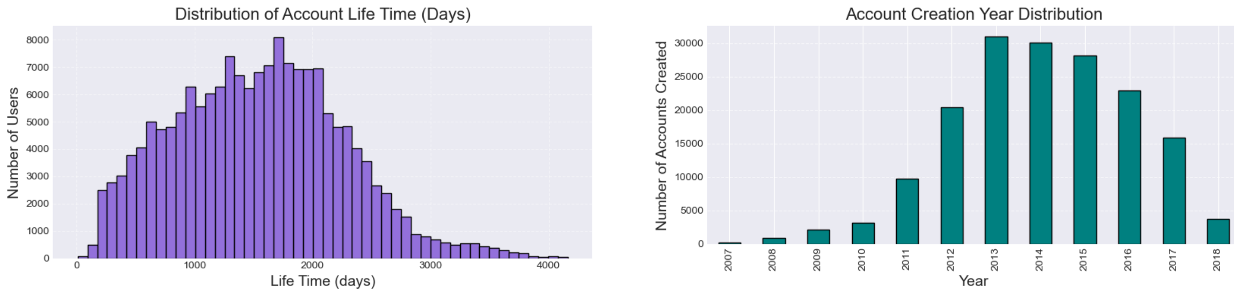


Figure 1: Language distribution among Twitch users.



(a) Distribution of user account lifetimes.

(b) Distribution of account creation years.

Figure 2: Temporal characteristics of Twitch users based on account metadata.

million edges. Metadata attributes include language, affiliate status, maturity flag, account lifetime, and account creation timestamps

To better understand the dataset, we conducted exploratory analysis on user attributes. As shown in Figure 1, the majority of Twitch users in the dataset are English-speaking, followed by smaller proportions of German, French, Spanish, and Russian speakers. This indicates a strong linguistic skew which could affect friendship prediction patterns. Figure 2a shows the distribution of user account lifespans. Most users have been active for over 1000 days, with a peak between 1500 and 2000 days, suggesting a relatively mature and stable user base. Finally, Figure 2b highlights account creation trends over time. The number of new accounts rose significantly from 2011 to 2013 and remained high until 2016, after which it gradually declined. This pattern may reflect Twitch's growth trajectory and should be considered when modeling temporal dynamics in the network.

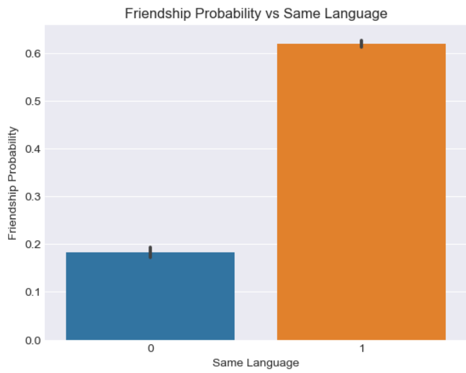
3.2 Pair Construction and Labeling

We randomly sampled 10,000 positive examples from the existing edge list to represent known friendships. Negative examples were generated by sampling random user pairs that are not connected in the graph, while ensuring they are valid users in the feature set. Each pair was labeled as 1 (friend) or 0 (non-friend).

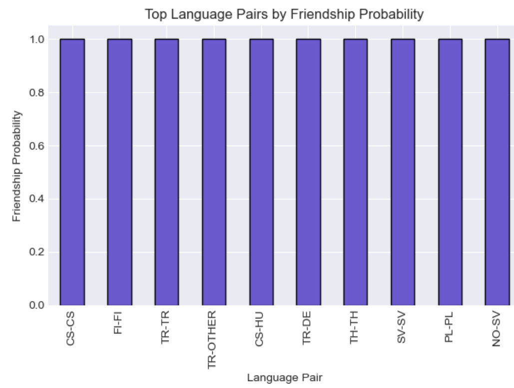
To better understand potential predictors of friendships, we examined linguistic features among pairs. As shown in Figure 3, user pairs who share the same language have a substantially higher probability of being friends (approximately 62%) compared to those who do not (around 18%). Additionally, the top-ranked language pairs (e.g., CS-CS, FI-FI, TR-TR) exhibit near-certain friendship probabilities, indicating a strong presence of linguistic homophily in Twitch's social structure.

3.3 Feature Engineering

Metadata Features: Each user pair was represented by several profile-level features designed to capture demographic and temporal similarities: `same_language`, a binary flag for matching languages; `same_affiliate` and `same_mature`, indicating shared affiliate status and mature content preference; `life_time_diff` and `avg_life_time`, representing the difference and average in account age; `creation_year_diff`, the absolute year difference; and



(a) Friendship probability vs. same language.



(b) Top language pairs by friendship probability.

Figure 3: Language similarity and pairwise friendship likelihood.

`is_same_year_created`, a flag for identical account creation years.

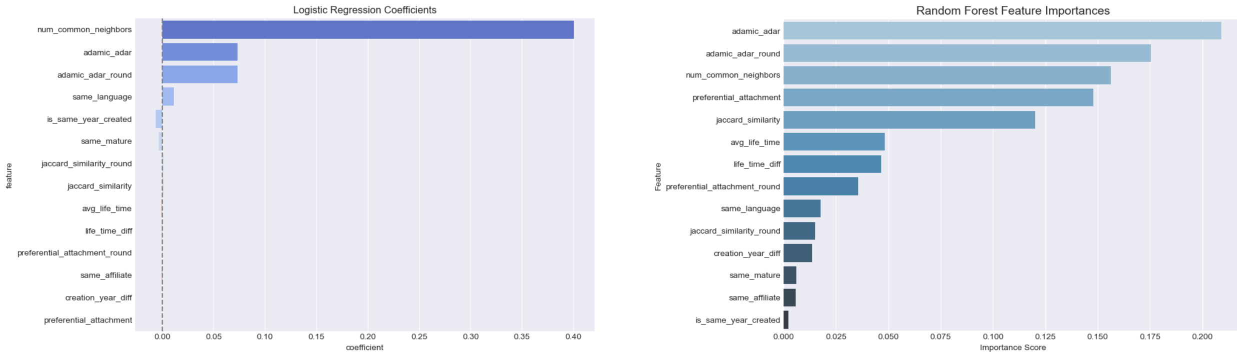
Graph-Based Features: We also constructed an undirected social graph using NetworkX and extracted key topological features like: `num_common_neighbors`- depicting the number of shared friends, `jaccard_similarity`- which is helpful in visualizing normalized neighbor overlap, `preferential_attachment`- which tells us the degree product, and `adamic_adar`- signifying a weighted score favoring rarer shared neighbors.

3.4 Modeling Approaches

We evaluated both metadata-only and full-feature models. For traditional classifiers, we used Random Forest and Logistic Regression. Models were trained using an 80/20 train-test split. We assessed performance using accuracy, F1-score, and AUC-ROC.

To understand which features most influenced the predictions, we analyzed the learned weights and importance scores from the two models. Figure 4 displays a comparison of feature importances for both Logistic Regression and Random Forest models.

In Logistic Regression, `adamic_adar` and `num_common_neighbors` possessed the highest positive coefficient values, reflecting strong linear association between these graph metrics and friendship probability. The Random Forest model, by contrast, strongly favored `adamic_adar` and `num_common_neighbors` again, and highly valued `preferential_attachment`, `jaccard_similarity`, reflecting that they can describe nonlinear relationships as well. Metadata features `account_age` and `same_language` contributed with lesser, but still significant, impact. This comparison shows that graph metrics are generally superior to user metadata as indicators of friendship and that ensemble models can more effectively leverage their intricate relationships.



(a) Logistic Regression Coefficients

(b) Random Forest Feature Importances

Figure 4: Comparison of feature influence across models. Logistic Regression highlights linear contributions, while Random Forest captures complex interactions.

3.5 Node2Vec Embeddings

We learned richer structural properties by training Node2Vec in a subgraph of top 10,000 high-degree nodes. We represented every user as a vector of length 64. We calculated the Hadamard product of embeddings of every pair of users to create a feature vector. We learned Random Forest, Logistic Regression, and XGBoost classifiers with these representations.

3.6 Evaluation Strategy

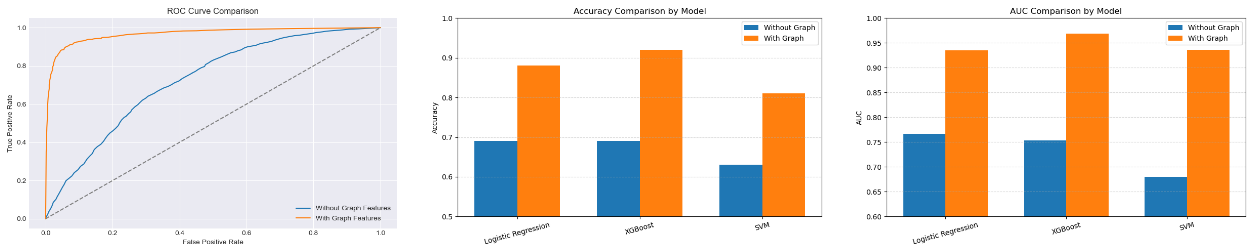
We compared models across four feature sets: metadata only, metadata plus graph features, Node2Vec embeddings, and Node2Vec combined with graph features. Models incorporating graph-based signals consistently outperformed those using metadata alone, underscoring the importance of structural context in friendship prediction.

4 Results

In this section, we report performance results of different models learned to forecast friendships between Twitch users. We compare models using different feature combinations—metadata only, metadata plus graph-based features, Node2Vec embeddings, and combining Node2Vec with graph-based features. As metrics of predictive ability, we utilize standard metrics like accuracy, F1-score, and Area Under the ROC Curve (AUC). Our experiment tries to find out what feature representations and modeling approaches provide maximum accuracy and give the best indication of the underlying structure of the Twitch network.

4.1 Graph Features vs. Metadata

To evaluate the contribution of graph-based features relative to metadata alone, we trained models under both settings and compared their performance using accuracy, AUC, and ROC curves.



(a) ROC Curve Comparison (b) Accuracy Comparison by Model (c) AUC Comparison by Model

Figure 5: Comparison of model performance using metadata only vs. metadata with graph features. Graph-enhanced models consistently outperform those without graph features across all evaluation metrics.

Figure 5 summarizes these comparisons across Logistic Regression, XGBoost, and SVM classifiers. The ROC curve in Figure 5a shows that models incorporating graph features (orange) achieve significantly higher true positive rates for nearly all thresholds. The accuracy comparison in Figure 5b shows that all models benefit from graph features, with the most substantial improvement observed in XGBoost. Similarly, the AUC comparison Figure 5c highlights significant gains in classification quality when graph-based features are included, confirming better model confidence and discrimination between friend and non-friend pairs.

These results demonstrate the predictive advantage of structural features in friendship inference. While metadata alone provides some signal, incorporating network topology captures relational patterns crucial for accurate predictions.

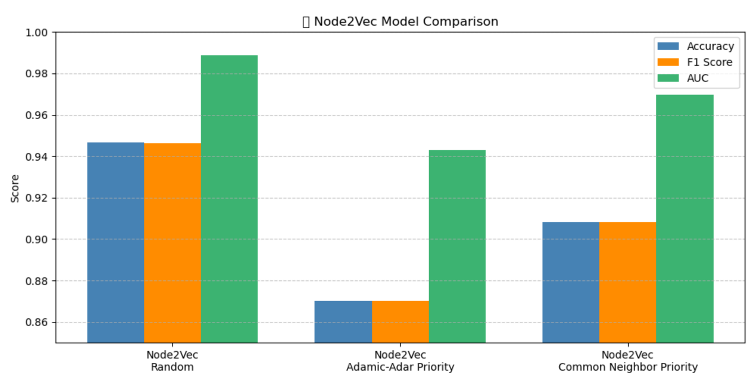


Figure 6: Node2Vec model comparison based on different sampling strategies.

4.2 Node2Vec Sampling Strategy

Due to memory constraints, it was not feasible to embed the entire Twitch graph. Instead, we sampled a subgraph of 3000 nodes and evaluated three sampling strategies: random sampling, Adamic-Adar priority sampling, and Common Neighbor priority sampling. Each configuration was averaged over 50 runs to ensure robustness, as shown in Figure 6.

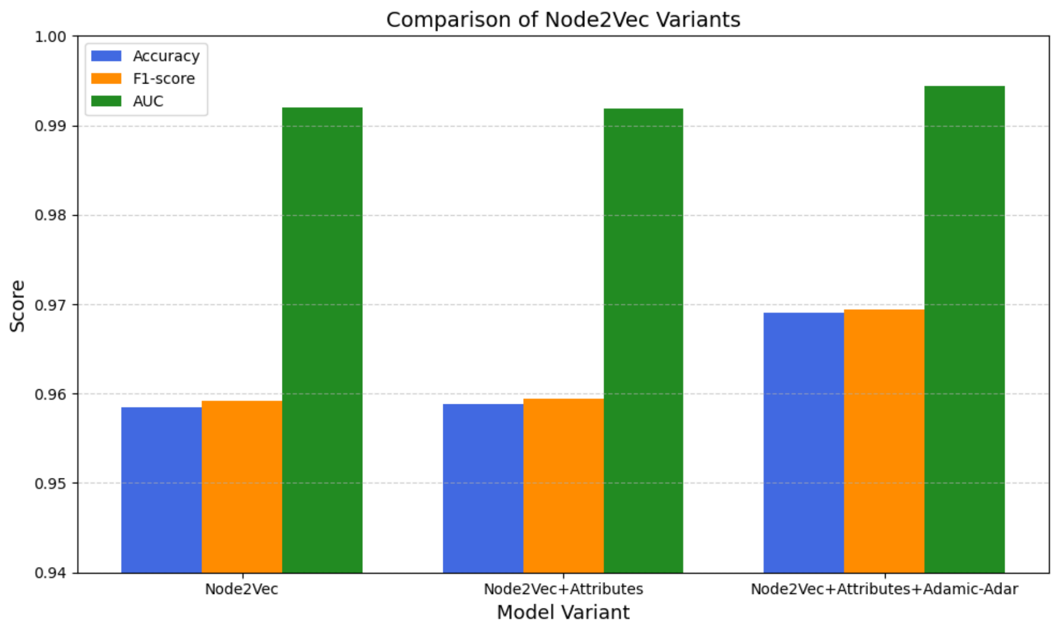


Figure 7: Comparison of Node2Vec variants. Combining embeddings with attributes and heuristics leads to improved classification.

Figure 6 illustrates relative results of accuracy, F1-score, and AUC. Of the algorithms tested, Node2Vec with random sampling produced the highest values overall, particularly in relation to AUC. Common neighbor priority sampling performance was also robust and less volatile than in the Adamic-Adar strategy, which trailed in every metric. The results of this suggest that priority-based sampling is promising, however, the structural distribution of sampled nodes strongly impacts embedding quality and downstream prediction.

4.3 Combined Models

We then investigated if combining Node2Vec embeddings with other features would provide higher prediction accuracy. We tested three variants specifically: using just Node2Vec, combining Node2Vec with metadata (user attributes), and combining Node2Vec with both metadata and the Adamic-Adar feature.

The results for these variants appear in Figure 7, and these were measured by accuracy, F1-score, and AUC. Node2Vec by itself performs well, and by combining it with attributes, we find there is a slight increase in accuracy and F1-score. The optimal performance occurs when combining Node2Vec, attributes, and Adamic-Adar, indicating that combining learned embeddings with profile features and interpretive heuristics has complementary advantages.

Table 1 gives an overall overview of the performance of all models tested. The Node2Vec + Metadata + Adamic-Adar combination proved to perform best among any combination, with the best result in all metrics, with accuracy above 0.969 and AUC above 0.994. This indicates that there exists good synergy between heuristic graph features and deep embeddings when used in conjunction with metadata.

Model	Accuracy	F1-score	AUC
Metadata Only	0.88	0.88	0.94
Metadata + Graph	0.93	0.93	0.97
Node2Vec Only	0.9585	0.9592	0.9920
Node2Vec + Metadata	0.9588	0.9594	0.9919
Node2Vec + +Metadata + Adamic-Adar	**0.9691**	**0.9694**	**0.9944**

Table 1: Performance of different models

These results emphasize that while Node2Vec provides robust representations, integrating additional contextual and structural information can further refine predictive capability.

5 Discussion

In this study, we investigated friendship prediction in Twitch's social network using both metadata and structural graph features. We tested both traditional machine learning and embedding-based approaches, examining how different types of features contribute to model accuracy. Although metadata-only models produced decent accuracy, additional integration of graph-based heuristics like common neighbors, Jaccard similarity, and Adamic-Adar greatly improved predictive accuracy using any of the tested classifiers. Embedding-based approaches with Node2Vec proved effective, and when integrated with metadata and graph heuristics, their performance improved even further. The optimal results were with Node2Vec and Adamic-Adar, with accuracy of 96.9% and AUC of 0.9944, demonstrating the power of hybrid feature approaches.

Even with strong results, we were constrained by computational needs to sample subgraphs and by working with static snapshots of a dynamic platform, which is Twitch. Potential future research could investigate dynamic link prediction with temporal graph embeddings [8] and recurrent models, or utilize more scalable approaches such as contrastive and graph neural networks. Smarter sampling and augmentation could also improve embedding quality with partial data [9]. Our results overall indicate that combining topological, embedding-based, and metadata features can form an effective backbone to construct personalized and scalable friend recommendation systems in big social platforms such as Twitch.

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Code Repository

The source code for this project is available at:
`https://github.com/nasifprangon/Twitch-Friendship-Prediction`